

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Winter 2023

Course: B. Tech

Branch: Electrical

Semester: IV

Subject Code & Name: Analog and Digital Electronics (BTEEE406B_Y19)

Max Marks: 60

Date: 29-01-24

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.

Marks

Q. 1 Solve Any Two of the following.

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| A) Explain small signal low frequency h-parameter model of transistor amplifiers and also define h-parameters. | 06 |
| B) Determine h-parameters from input and output characteristics of common base transistor. | 06 |
| C) Explain RC coupled amplifier with its circuit diagram | 06 |

Q.2 Solve Any Two of the following.

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| A) Draw and explain non-inverting amplifier with output equation. | 06 |
| B) With a neat circuit diagram show that an Op-Amp could be used as a summer. | 06 |
| C) Explain Op-Amp as a differentiator with neat circuit diagram and necessary equations. | 06 |

Q. 3 Solve Any Two of the following.

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| A) Convert the following numbers:
a) $(0.1101)_2 = (\quad ? \quad)_{10}$
b) $(12.4)_{10} = (\quad ? \quad)_8$
c) $(FF12)_{16} = (\quad ? \quad)_{10}$ | 06 |
| B) Explain all logic gates with their symbols and truth tables. | 06 |
| C) Prove the following De Morgan's theorems:
a) $AB + CD = \overline{\overline{A}\overline{B}.\overline{C}\overline{D}}$
b) $(A + B).(C + D) = \overline{\overline{(A + B)} + \overline{(C + D)}}$ | 06 |

Q.4 Solve Any Two of the following.

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| A) What are different operational properties of logic family? Explain any 3 properties in detail. | 06 |
| B) Draw and explain 2-input NMOS NAND and NOR logic gates. | 06 |
| C) Design an S-R latch using 2-inputs NOR gates. | 06 |

Q. 5 Solve Any Two of the following.

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| A) Minimize the following expression using K-maps and realize using NOR gates only.
$f_1(A, B, C, D) = \pi M(1, 2, 3, 5, 6, 7, 9, 10, 11, 13, 14, 15)$ | 06 |
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B) Explain half adder circuit with the truth table. **06**

C) Minimize the following logic functions and realize using NAND/NOR gates. **06**

$$f_1(A, B, C, D) = \sum m(1, 3, 5, 8, 9, 11, 15) + d(2, 13)$$

***** End *****